# Inequalities of Real Numbers.

## Bernoulli's Inequality

If $a>-1$, $1+na\le(1+a)^n$ for all $n\in\mathbb{N}$.

Prof. If $n=1$, trivial. If the statement holds for $n=k\in\mathbb{N}$, then

$$(1+a)^{k+1}=(1+a)(1+a)^k\underset{1+a\ge 0}{\ge}(1+a)(1+ka)=1+(k+1)a+a^2\ge 1+(k+1)a.$$

## Arithmetic-Geometric mean

If $a,b\in\mathbb{R}$, then $2ab\le a^2+b^2$ (Directly, it can be proved by $(a-b)^2\ge 0$.)

## Generalized A-G mean inequality,

If $a_i\ge 0$, then for all $n\in\mathbb{N}$,

$$(a_1\cdot a_2\cdots a_n)^{\frac{1}{n}}\le\frac{1}{n}\cdot(a_1+a_2+\cdots+a_n)$$

prof. If $n=2^1$, $\sqrt{a_1a_2}\le\frac{1}{2}(a_1+a_2)$, thus the statement holds.

Suppose that for $n=2^k$ the statement be hold. Then is

$$(a_1\cdots a_n)^{\frac{1}{n}}\le\frac{1}{n}(a_1+\cdots+a_n)$$

Now, $\frac{1}{2n}(a_1+\cdots+a_{2n})=\frac{1}{2}\left(\frac{1}{n}(a_1+\cdots+a_n)+\frac{1}{n}(a_{n+1}+\cdots+a_{2n})\right)$

$$\ge\frac{1}{2}\left((a_1\cdots a_n)^{\frac{1}{n}}+(a_{n+1}\cdots a_{2n})^{\frac{1}{n}}\right)$$

$$=\frac{1}{2}\left(\left((a_1\cdots a_n)^{\frac{1}{2n}}\right)^2+\left((a_{n+1}\cdots a_{2n})^{\frac{1}{2n}}\right)^2\right)$$

$$\ge\frac{1}{2}\cdot 2(a_1\cdots a_n)^{\frac{1}{2n}}\cdot(a_{n+1}\cdots a_{2n})^{\frac{1}{2n}}$$

$$=(a_1\cdots a_n a_{n+1}\cdots a_{2n})^{\frac{1}{2n}}$$

Therefore, by induction, the statement hold when $n=2^k$ $(k\in\mathbb{N})$.

To generalize, let $n\ge 2$ be given. Put $k\in\mathbb{N}$ s.t.

$$2^{k-1}\le n\le 2^k \qquad \frac{1}{2^k}\le\frac{1}{n}\le\frac{1}{2^{k-1}} \qquad 1\le 2$$

Put

$$b_i=\begin{cases}a_i & \text{if } i\le n\\ \frac{1}{n}(a_1+\cdots+a_n)=:M & \text{if } n<i\le 2^k\end{cases} \qquad \frac{n}{2^k}\le 1 \qquad 0\le 1-\frac{n}{2^k}$$

Then, $(b_1\cdots b_{2^k})^{\frac{1}{2^k}}\le\frac{1}{2^k}\cdot(b_1+\cdots+b_{2^k})=\frac{1}{2^k}\cdot(nM+(2^k-n)M)=M$

And $(b_1\cdots b_{2^k})^{\frac{1}{2^k}}=(a_1\cdots a_n\cdot M^{2^k-n})^{\frac{1}{2^k}}=(a_1\cdots a_n)^{\frac{1}{2^k}}\cdot M^{(1-\frac{n}{2^k})}$

Finally, $(a_1\cdots a_n)^{\frac{1}{2^k}}\le M^{\frac{n}{2^k}}$ implies $(a_1\cdots a_n)^{\frac{1}{n}}\le(a_1+\cdots+a_n)\cdot\frac{1}{n}$.

「Lagrange's Inequality.
Let $a_i, b_i \in \mathbb{R}$. Then,

$$\left[\sum_{i=1}^{n} a_i b_i\right]^2 = \left[\sum_{i=1}^{n} a_i^2\right] \cdot \left[\sum_{i=1}^{n} b_i^2\right] - \frac{1}{2} \cdot \sum_{1 \le j,k \le n} (a_j b_k - a_k b_j)^2$$